

Liam C Pendleton, MSc

Quantitative Biologist

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Summary

Avian conservation scientist with 8+ years of experience leveraging analytical and field methods to inform conservation decisions. I use long-term ecological monitoring, predictive models, and machine learning methods to monitor avian population trends and behavior. My expertise includes frequentist and Bayesian statistical modeling, data management, spatial and machine learning analytical methods, team supervision, and cross-sector collaboration.

Skills

Data Analytics & Management • Statistical Modeling • Machine Learning • Data Visualization • Geospatial Analysis • GIS • GitHub • Data Management • SQL • R • Technical Writing • Project Management • Community Outreach • Field Work • Cross-Sector Collaboration • Habitat assessments • Vegetation Surveys • Permitting & Regulatory Compliance

Professional Experience

Washington Cooperative Fish and Wildlife Research Unit

Seattle, WA

Research Assistant

07/2021 - 06/2024

Collaborated with state, federal and community science partners to design and implement a long-term seabird monitoring program, using Bayesian modeling, GIS, and machine learning techniques to inform population trends and behavior of two seabird species monitored by the Puget Sound Partnership as indicators of Puget Sound restoration goals.

- Transformed an unstructured, multi-decade time series into a relational database, improving data accessibility and reducing processing time
- Built hierarchical models in a Bayesian framework using R and JAGS to estimate effects of biological and physical processes on reproductive success of pigeon guillemots
- Used hidden Markov modeling in R to identify state-specific behaviors and foraging habitat of pigeon guillemots and rhinoceros auklets
- Calibrated and issued full-spectrum acoustic loggers to monitor and describe seabird species diversity on a remote Pacific atoll
- Organized and trained field teams of 5+ members annually, and directed 20+ field days per season to collect spatial and reproductive data
- Communicated research to audiences of varying levels of expertise at conferences, symposia and open house events

Bird Laboratory

Ypsilanti, MI

Research Assistant

08/2019 - 05/2021

Designed and implemented a spatial analysis study on a protected songbird species, using R-based modeling and tracking data to define and identify core habitat, highlighting potential conservation priorities across 5,000+ hectares of habitat.

- Managed a large-scale spatial dataset describing habitat usage across 5,000+ hectares; validated and standardized data to support analyses
- Used kernel density estimations and Brownian bridge movement models in R to describe core habitats and foraging patterns
- Analyzed acoustic files using Audacity to assess energy expenditure of breeding and non-breeding red crossbills
- Led logistical planning for a multi-week field data collection trip along the US west coast, including travel coordination, equipment procurement, and scheduling for team members
- Collected morphometric and physiological data from 12 birds, adding ~20% more datapoints to existing datasets
- Presented research at multiple conferences; communicated complex scientific concepts appropriately to diverse audiences

Aquatic Geochemistry Laboratory

Ann Arbor, MI

Data Analyst

02/2020 - 09/2020

Analyzed water quality from 50+ Alaskan permafrost samples, identifying key chemical drivers of carbon conversion while streamlining research workflows through statistical modeling, version control, and reporting.

- Used R and Excel to assess annual variation in 15+ chemical properties of 50+ Alaskan permafrost samples, identifying sunlight, bacterial activity and iron concentration as significant contributors to conversion into greenhouse gases
- Provided weekly data summaries to the principal investigator, resulting in improved efficiency of ongoing lab projects and project workflow
- Used GitHub to track and manage 5 iterations of analysis code, ensuring full version control

Professional Experience

Bird Center of Washtenaw County Data Coordinator & Supervisor

Ann Arbor, MI
05/2017 - 11/2019

Oversaw patient data management and tracking, leading database optimization, digital system implementation, and data-driven decision-making to improve operational efficiency and patient care.

- Managed a database containing 3,000+ patient records, overseeing data entry and retrieval processes
- Led a transition from paper-based records to a digital patient management system and implemented a standardized data collection protocol
- Wrote annual reports describing clinic statistics including seasonal trends in patient circumstances, rehab success rates, and species treated
- Supervised and trained 20+ interns and volunteers, improving efficiency in wildlife rehabilitation practices
- Coordinated annual fundraising and community outreach events, securing \$30,000+ in donations each year to support wildlife care operations
- Collaborated with board members and external partners to build partnerships supporting the Bird Center's growth

Detroit Audubon Society Field Technician

Detroit, MI
05/2018 - 08/2019

- Monitored 30+ shorebird nests across multiple sites, using handheld GPS to track and report sites and environmental conditions
- Collected and analyzed aquatic condition data (e.g., water levels, temperature, vegetation types)
- Assisted in restoring 3 acres of native prairie habitat using vegetation surveys and habitat assessments, increasing nesting and foraging opportunities for native bird and pollinator species
- Led community outreach efforts, engaging with 50+ residents to promote habitat conservation and increase awareness of native wildlife

Education

University of Washington M.S., Aquatic and Fishery Sciences

Seattle, WA | 2024

Eastern Michigan University B.S., Biology

Ypsilanti, MI | 2021

Washtenaw Community College A.A., Liberal Arts

Ann Arbor, MI | 2018

Volunteering

Pacific Seabird Group Organizer

Seattle, WA
11/2023 - 03/2024

- Designed and ran a professional development panel targeted at early career scientists. Contacted and organized panelists from varying professional backgrounds, wrote starter questions, and facilitated subsequent dialogue between panelists and audience.
- Assisted in organizing a professional mentoring event. Contacted and organized candidate mentors, and paired student mentees with mentors.

Salish Sea Guillemot Network Field Technician

Seattle, WA
05/2022 - 08/2023

- Monitored foraging activity of pigeon guillemots at various sites around the Puget Sound, WA. Identified prey species, nest locations, and provisioning frequency.

Ann Arbor Natural Area Preservation Field Technician

Ann Arbor, MI
05/2019 - 08/2020

- Conducted point counts and vegetation surveys at 10+ sites around Ann Arbor, MI to assess songbird abundance and diversity following habitat restoration efforts.